

A 3×3 COUNTEREXAMPLE FAMILY TO CONJECTURE 4.5 OF CHEN–CAO’S ARXIV:1801.00225V1

ABSTRACT. Let ω_n be the set of $n \times n$ doubly substochastic matrices (nonnegative, all row sums and all column sums at most 1), let $\sigma(A)$ be the sum of all entries of A , and let $\omega_n^s = \{A \in \omega_n : \sigma(A) = s\}$. Chen and Cao determined $\max\{\text{per}(I - A) : A \in \omega_n^s\}$ for n even and for n odd with $s \leq n - 1$, left the remaining cell open, and conjectured a two-branch value for $n = 3$, $2 < s \leq 3$ (their Conjecture 4.5), whose printed form needs two readings fixed before it can be tested (Section 1). In the fixed-mass reading, with the second branch read as the one their own Lemma 4.4 proves, the conjecture is false at *every* s in the open interval $(2, 3)$. The witness is the one-parameter family

$$A(s) = \begin{pmatrix} 0 & 1-u & u \\ 1-u & 0 & u \\ u & u & 0 \end{pmatrix}, \quad u = \frac{s-2}{2},$$

$$\text{per}(I - A(s)) = \frac{s^3 - 5s^2 + 4s + 12}{4},$$

which lies in ω_3^s and exceeds *both* of the conjectured branches, the excess over the first being exactly $s(s-3)^2/4$. At $s = 5/2$ the conjecture predicts $23/16$ and the witness gives $51/32$. Equivalently, Chen–Cao’s own Lemma 4.4 is not tight anywhere on $(2, 3)$. The true value of $\max\{\text{per}(I - A) : A \in \omega_3^s\}$ remains open; at $s = 5/2$ what is established here is $51/32 \leq \max \leq 2$. The journal version of record was not obtained, so every statement here about what the source prints is a statement about the e-print [arXiv:1801.00225v1](#).

1. THE STATEMENT REFUTED

Notation, in the source’s own words. An $n \times n$ nonnegative matrix is *doubly stochastic* if every row sum and every column sum equals 1, and *doubly substochastic* if every row sum and every column sum is at most 1. Write Ω_n for the doubly stochastic matrices, ω_n for the doubly substochastic ones, and $\tilde{\omega}_n$ for the *row* substochastic ones (only the row sums are constrained). For a square matrix $A = [a_{ij}]$ let $\sigma(A) = \sum_{i,j} a_{ij}$, and put

$$\omega_n^s = \{A \in \omega_n : \sigma(A) = s\}, \quad \tilde{\omega}_n^s = \{A \in \tilde{\omega}_n : \sigma(A) = s\}.$$

No symmetry, no zero diagonal and no rationality is assumed anywhere in ω_n . The distinction between ω_n and $\tilde{\omega}_n$ is the axis of this note: Chen and Cao settled the row substochastic problem in the cell at issue and left the doubly substochastic one open.

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The source statement, verbatim. Chen and Cao [1] prove

$$(1) \quad \max\{\text{per}(I - A) : A \in \omega_n^s\} = 2^{e/2} \left[1 + \left(\frac{s-e}{2} \right)^2 \right], \quad e = 2\lfloor s/2 \rfloor,$$

under the hypothesis “ n is even, or n is odd and $\sigma(A) \leq n - 1$ ” (their Theorem 3.1; their Theorem 3.13 is the same formula over $\tilde{\omega}_n^s$), and they say of the excluded cell that “for $A \in \omega_n^s$, the upper bound of $\text{per}(I - A)$ still remains mystery”. In Section 4 they settle the row substochastic half of that cell,

$$(2) \quad \max\{\text{per}(I - A) : A \in \tilde{\omega}_n^s, n - 1 < s \leq n\} = 2^{\frac{n-1}{2}} \quad (n \text{ odd}),$$

observe that their maximiser there is *not* doubly substochastic (a column sum exceeds 1), and then conjecture the $n = 3$ value.

All statement numbers and all quotations below are from the L^AT_EX e-print arXiv:1801.00225v1 (<https://arxiv.org/e-print/1801.00225>), whose single source file `per_I-A_LAASubmission.tex` has SHA-256 beginning 9113694030113dda; the statement numbering follows from its preamble, in which `thm`, `lem` and `conj` share one counter. The target statement, verbatim:

$$(3) \quad \max_{A \in \omega_3} \{\text{per}(I - A)\} = \begin{cases} \frac{\sigma^2(A) - 5\sigma(A) + 12}{4} & \text{if } \frac{-3 + \sqrt{57}}{2} < \sigma(A) \leq 3, \\ 6 - 4\sigma(A) & \text{if } 2 < \sigma(A) \leq \frac{-3 + \sqrt{57}}{2}, \end{cases}$$

and the half of it the authors *prove*, verbatim (Lemma 4.4):

(4)

$$\max_{A \in \omega_3} \{\text{per}(I - A)\} \geq \max \left\{ \frac{\sigma^2(A) - 5\sigma(A) + 12}{4}, 6 - 2\sigma(A) \right\} \quad (2 < \sigma(A) \leq 3),$$

proved by exhibiting the two matrices

$$(5) \quad A_0 = \begin{pmatrix} 0 & \frac{1}{2} & \frac{\sigma}{2} - 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{\sigma}{2} - 1 & \frac{1}{2} & 0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \sigma - 2 \end{pmatrix},$$

with $\text{per}(I - A_0) = \frac{\sigma^2 - 5\sigma + 12}{4}$ and $\text{per}(I - A_1) = 6 - 2\sigma$. **So (3) is precisely the assertion that (4) is tight**, and that is what we refute.

Two readings, and which one is refuted. Two points of reading must be fixed before anything is claimed.

First, (3) is printed as a maximum over all of ω_3 whose right-hand side depends on $\sigma(A)$; read literally that is ill-formed, since $\max_{A \in \omega_3} \text{per}(I - A)$ is the single number 2. The intended *fixed-mass* reading, max over ω_3^σ , is forced by (4) directly above it and by Conjecture 4.6, which writes $\max\{\text{per}(I - B) : B \in \omega_3^{s'}\}$ explicitly. That is the reading refuted here.

Second, the printed second branch $6 - 4\sigma$ is negative for $\sigma > 3/2$ and hence throughout its own stated range, whereas $\text{per}(I - A) \geq \det(I - A) \geq 0$ for doubly substochastic A ; it is also contradicted in print by (4), which the

source states just above it. It is a misprint for the $6 - 2\sigma$ of (4), and three independent arguments say so: the printed threshold $\frac{-3+\sqrt{57}}{2}$ is the root of $s^2 + 3s - 12$, i.e. the crossover of the first branch with $6 - 2s$ (the crossover with $6 - 4s$ is the root of $s^2 + 11s - 12$, namely $s = 1$, outside $(2, 3]$ entirely); the authors' own A_1 realises $6 - 2\sigma$ exactly and $6 - 4\sigma$ never; and positivity forbids $6 - 4\sigma$. **Refuting the printed reading would therefore be worth nothing, and we do not claim it as content.** We refute the corrected, and strongest, reading

$$(6) \quad V(s) := \max\left\{\frac{s^2-5s+12}{4}, 6-2s\right\},$$

which is exactly the lower bound (4) proves. As it happens the witness below exceeds $6 - 4s$ as well, so the refutation is independent of how the misprint is read; and at the headline mass $s = 5/2$ the governing branch is the first one, printed identically in the *proved* (4), so there the two readings coincide.

What text we read. The version of record was not obtained. Every sentence of this note about “what the source prints” is a statement about arXiv:1801.00225v1 and nothing else. The arXiv abstract page carries [v1] only and no journal-ref; we found no open copy of the journal version; and the reference list deposited with doi:10.1016/j.laa.2018.06.031 agrees with the preprint’s twelve \bibitems in eleven entries and differs in one, a term-rank paper replacing a preprint citation, so refereeing did alter the document. The Crossref record for the DOI carries no update or relation entry and we found no corrigendum notice; we do not conclude from that that no correction exists, since the journal text itself was not read and one citation-index channel could not be read either. We therefore state the claim in the form: *Conjecture 4.5 of arXiv:1801.00225v1, in the reading fixed above, is false.* The residual risk is discussed in Section 4.

2. THE WITNESS

For $s \in (2, 3)$ set $u = \frac{s-2}{2} \in (0, \frac{1}{2})$ and

$$(7) \quad A(s) = \begin{pmatrix} 0 & 1-u & u \\ 1-u & 0 & u \\ u & u & 0 \end{pmatrix}.$$

Membership in ω_3^s is checkable by eye. The entries are 0, $u > 0$ and $1-u > 0$. The row sums are $(1, 1, 2u)$ and, since $A(s)$ is symmetric, the column sums are the same; $2u = s - 2 \leq 1$ exactly because $s \leq 3$. Finally $\sigma(A(s)) = 2(1-u) + 4u = 2 + 2u = s$, exactly. The column-sum clause is the one that matters: dropping it would place the matrix in the row substochastic class $\tilde{\omega}_3$, where the maximum (2) is already known.

For a zero-diagonal $A = \begin{pmatrix} 0 & a & b \\ c & 0 & d \\ e & f & 0 \end{pmatrix}$ the six permutations of S_3 give

$$(8) \quad \text{per}(I - A) = 1 + ac + be + df - ade - bcf,$$

so with $a = c = 1 - u$ and $b = d = e = f = u$,

$$(9) \quad \begin{aligned} \text{per}(I - A(s)) &= 1 + (1 - u)^2 + 2u^2 - 2(1 - u)u^2 = 2 - 2u + u^2 + 2u^3 \\ &=: f(s) = \frac{s^3 - 5s^2 + 4s + 12}{4}. \end{aligned}$$

The headline instance. At $s = 5/2$, i.e. $u = 1/4$,

$$A = \begin{pmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & 0 \end{pmatrix} \in \omega_3, \quad \sigma(A) = \frac{5}{2},$$

with row sums and column sums $(1, 1, \frac{1}{2})$ and

$$\text{per}(I - A) = 1 + \frac{9}{16} + \frac{1}{16} + \frac{1}{16} - \frac{3}{64} - \frac{3}{64} = \frac{51}{32} = 1.59375.$$

Since $\frac{5}{2} > \frac{-3+\sqrt{57}}{2} = 2.274917\dots$, the first branch of (3) is in force and predicts $\frac{25/4-25/2+12}{4} = \frac{23}{16} = 1.4375$. The excess is exactly $\frac{5}{32}$ and the ratio is $\frac{51}{46}$.

Exact values. Table 1 lists every value we record. All entries are exact rationals.

s	$1 - u, u$	$\text{per}(I - A(s)) = f(s)$	$V(s)$	branch	excess
21/10	19/20, 1/20	7611/4000 = 1.902750	9/5 = 1.8	second	411/4000
9/4	7/8, 1/8	453/256 = 1.769531	3/2 = 1.5	second	69/256
5/2	3/4, 1/4	51/32 = 1.593750	23/16 = 1.4375	first	5/32
11/4	5/8, 3/8	383/256 = 1.496094	93/64 = 1.453125	first	11/256
14/5	3/5, 2/5	186/125 = 1.488000	73/50 = 1.46	first	7/250
29/10	11/20, 9/20	5939/4000 = 1.484750	591/400 = 1.4775	first	29/4000
2	1, 0	2	2	—	0
3	1/2, 1/2	3/2	3/2	—	0

TABLE 1. The witness against the corrected conjectured value (6). In the last two rows the mass is an endpoint, where the true maximum is published and the family returns exactly it, with no violation: $f(2) = 2$ is (1) at $n = 3$, $e = 2$, and $f(3) = 3/2$ is Chen–Cao’s Lemma 4.2, the value over Ω_3 .

3. THE REFUTATION

Theorem 1. *For every s with $2 < s < 3$ the matrix $A(s)$ of (7) lies in ω_3^s and*

$$\max\{\text{per}(I - A) : A \in \omega_3^s\} \geq f(s) > V(s),$$

where V is the value (6): Conjecture 4.5 of `arXiv:1801.00225v1` in the reading fixed in Section 1, which is also the lower bound Lemma 4.4 proves. Consequently Conjecture 4.5 of `arXiv:1801.00225v1`, in the reading fixed

in Section 1, is false at every mass in $(2, 3)$, and Lemma 4.4 is not tight anywhere on that interval.

Proof. Membership and the value $f(s)$ are Section 2. Two polynomial identities finish it; both are exact coefficient identities, not numerical comparisons. First,

$$(10) \quad f(s) - \frac{s^2 - 5s + 12}{4} = \frac{s^3 - 6s^2 + 9s}{4} = \frac{s(s-3)^2}{4} > 0 \quad (2 < s < 3),$$

since $s > 0$ and $s \neq 3$. Second, with $t = s - 2 \in (0, 1)$,

$$(11) \quad f(s) - (6 - 2s) = \frac{s^3 - 5s^2 + 12s - 12}{4} = t + \frac{t^2}{4} + \frac{t^3}{4} > 0.$$

Because V is the *maximum* of the two branches and f exceeds both of them at every interior mass, $f > V$ there. (For completeness, f also exceeds the printed branch $6 - 4s$, which is negative on the whole interval, so no reading of the misprint and no reading of the threshold rescues the conjecture.) Equality in (10) occurs on $[2, 3]$ only at $s = 3$, and (11) degenerates only at $s = 2$. $\square$

The witness respects the two upper bounds proved for this functional: since $A(s) \in \omega_3 \subset \tilde{\omega}_3$, Chen–Cao’s (2) and Malek’s theorem [2] both give $\text{per}(I - A(s)) \leq 2$ at $n = 3$, and indeed $2 - f(s) = (s - 2)(2 + 3s - s^2)/4 > 0$ on $(2, 3]$, the roots of $2 + 3s - s^2$ being $\frac{3 \pm \sqrt{17}}{2}$.

4. WHAT IS NOT SETTLED

The upper bound is open, and this is the main limit. What is proved here is $\max \geq f(s)$ together with $f(s) > V(s)$. It is *not* proved that $\max = f(s)$. Malek’s bound [2] gives only $\max \leq 2$ at $n = 3$, and $f < 2$ on $(2, 3]$, so it does not decide the question. **At $s = 5/2$ the honest bracket is $51/32 \leq \max \leq 2$; the conjectured $23/16$ is excluded, and nothing here determines the true value.**

No optimality of $A(s)$ is claimed. $A(s)$ is exhibited only as a feasible point beating $V(s)$. **It is not claimed here to maximise $\text{per}(I - A)$ over ω_3^s ,** on any slice of that polytope or over all of it; no global search over the nine free entries is offered here, and none is performed by the program accompanying this note. The natural tool for a rigorous finish is Cao and Chen’s partition of the polytope of doubly substochastic matrices by the sum of all entries, which characterises the extreme points of exactly ω_n^s [3]. That $A(s)$ has a zero diagonal is a fact about it; nothing here assumes a zero diagonal, and Lemma 4.4’s own second witness A_1 of (5) carries $a_{33} = \sigma - 2 \neq 0$.

The residual risk in the journal text. Only arXiv:1801.00225v1 was read, and the reference list deposited with the DOI shows that it is not the printed document (Section 1). The refutation is unaffected by an edit to the misprinted branch or to the threshold, since at $s = 5/2$ the printed and the corrected readings coincide and the governing first branch there is printed identically in the *proved* (4). The risk that remains is wholesale replacement of the conjecture in print: if refereeing replaced the two-branch value by the cubic f , then this note shrinks to a correction of the preprint record. The refutation of the v1 text is unaffected either way.

Prior art. The source has four indexed citers, all of which were read; none of them quotes Conjecture 4.5 or 4.6 at all, they cite Lemma 4.2 and Theorems 3.1, 3.13 and 4.1. Malek’s theorem [2], which we read only through its zbMATH review, lands adjacent — an upper bound of $2^{\lfloor n/2 \rfloor}$, with no prescribed-mass parameter — rather than on the target. The citer list is not certified complete: one citation-index channel could not be read, and the zbMATH review of the source’s own record is licence-blocked; neither is read here as a negative.

The closest live work is Lavi [4], which settles a *neighbouring* conjecture of the same source — the Kim–Roush value $\max_{A \in \Omega_{2k+1}} \text{per}(I - A) = 3 \cdot 2^{k-2}$, i.e. the doubly stochastic endpoint $s = n$ — with an equality classification. It contains no counterexample to Conjecture 4.5 and no value on ω_3^s for $2 < s < 3$; it records the cell at issue here as open: “For odd n and $n - 1 < s \leq n$, they determined the row-substochastic maximum but left the doubly substochastic case open.” Two further papers by overlapping authors are adjacent but do not touch the target: Chen, Li, Yang, Zhu and Cao [5] prove convexity of the permanent along the segment $[I, A]$ for *row* substochastic A and conjecture a minimum over transportation polytopes, with no maximum over ω_n^s ; and Chen, Zhu, Li, Yang and Cao [6] characterise the extreme points of transportation polytopes with fixed total sum and give the maximum of $\text{per}(A)$ there, not of $\text{per}(I - A)$.

5. REPRODUCTION AND VERIFICATION

The load-bearing verification is by hand. Section 2 exhibits a 3×3 matrix with entries in $\{0, \frac{1}{4}, \frac{3}{4}\}$, six row and column sums, and a six-term permanent; Section 3 is two polynomial identities in one variable. No computation is needed to check any claim in this note, and none of the claims rests on floating-point arithmetic.

A program accompanies this note as a re-derivation, not as evidence: it recomputes every permanent by the full $n!$ expansion over exact rationals, reproduces Table 1 including the governing branch at each mass, and verifies (10) and (11) by exact coefficient comparison rather than by sampling. It also carries out checks on material not stated in this note — the geometry of the symmetric zero-diagonal slice at each tabulated mass, and a direct-sum

lift of the witness to $n = 5, 7$ — which support no claim made above. Its transcript records 97 passing checks and no failures.

What that run does and does not establish should be stated plainly. Its per-matrix checks — membership in ω_3 , the mass and the permanent — are carried out at the eight masses of Table 1, six interior and two endpoints, and at no other mass; the strict comparison $f(s) > V(s)$ is checked at the six interior masses only, while at $s = 2$ and $s = 3$ the program checks and records the equality $f(s) = V(s)$, with excess exactly 0 and no violation. Several of its check names and step headings are phrased as statements over the whole open interval, but what is evaluated at those places is exactly those sampled masses. The claim for every $s \in (2, 3)$ rests on the two polynomial identities of Section 3, which are proved by hand and re-checked by the program as identities between coefficients rather than at sample points; the strict signs of those two expressions on $(2, 3)$ are read off by hand and are not machine-verified. The program does *not* attempt the open upper bound of Section 4, performs no global search, contains no floating-point decision, and reads no source text: what the source prints is quoted from `arXiv:1801.00225v1` in Section 1, and the journal version of record was never obtained.

REFERENCES

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